

```
File Edit View Search Terminal Tabs Help
surabhi@surabhi-seng: /home/surabhi/Documents/Documents/Software/TensorFlo... x surabhi@surabhi-seng: /home/surabhi/Documents/Documents/Software/TensorFlo... x
[ OK ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_faster_rcnn_with_matmul
[ RUN ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_faster_rcnn_without_matmul
[ OK ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_faster_rcnn_without_matmul
[ RUN ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_mask_rcnn_with_matmul
[ OK ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_mask_rcnn_with_matmul
[ RUN ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_mask_rcnn_without_matmul
[ OK ] ModelBuilderTest.test_create_faster_rcnn_models_from_config_mask_rcnn_without_matmul
[ RUN ] ModelBuilderTest.test_create_rfcn_model_from_config
[ OK ] ModelBuilderTest.test_create_rfcn_model_from_config
[ RUN ] ModelBuilderTest.test_create_ssd_fpn_model_from_config
[ OK ] ModelBuilderTest.test_create_ssd_fpn_model_from_config
[ RUN ] ModelBuilderTest.test_create_ssd_models_from_config
[ OK ] ModelBuilderTest.test_create_ssd_models_from_config
[ RUN ] ModelBuilderTest.test_invalid_faster_rcnn_batchnorm_update
[ OK ] ModelBuilderTest.test_invalid_faster_rcnn_batchnorm_update
[ RUN ] ModelBuilderTest.test_invalid_first_stage_nms_iou_threshold
[ OK ] ModelBuilderTest.test_invalid_first_stage_nms_iou_threshold
[ RUN ] ModelBuilderTest.test_invalid_model_config_proto
[ OK ] ModelBuilderTest.test_invalid_model_config_proto
[ RUN ] ModelBuilderTest.test_invalid_second_stage_batch_size
[ OK ] ModelBuilderTest.test_invalid_second_stage_batch_size
[ RUN ] ModelBuilderTest.test_session
[ SKIPPED ] ModelBuilderTest.test_session
[ RUN ] ModelBuilderTest.test_unknown_faster_rcnn_feature_extractor
[ OK ] ModelBuilderTest.test_unknown_faster_rcnn_feature_extractor
[ RUN ] ModelBuilderTest.test_unknown_meta_architecture
[ OK ] ModelBuilderTest.test_unknown_meta_architecture
[ RUN ] ModelBuilderTest.test_unknown_ssd_feature_extractor
[ OK ] ModelBuilderTest.test_unknown_ssd_feature_extractor
-----
Ran 16 tests in 0.162s
OK (skipped=1)
surabhi@surabhi-seng: /home/surabhi/Documents/Documents/Software/TensorFlow/tensorflow/models/research$
```

Figure 2: Testing the installation

```
Object Detection Demo
Welcome to the object detection inference walkthrough! This notebook will walk you step by step through the process of using a pre-trained model to detect objects in an image. Make sure to follow the installation instructions before you start.

Imports
In [23]: 1 import numpy as np
          2 import os
          3 import six.moves.urllib as urllib
          4 import sys
          5 import tarfile
          6 import tensorflow as tf
          7 import zipfile
          8
          9 from distutils.version import StrictVersion
         10 from collections import defaultdict
         11 from io import StringIO
         12 from matplotlib import pyplot as plt
         13 from PIL import Image
         14
         15 # This is needed since the notebook is stored in the object_detection folder.
         16 sys.path.append("..")
         17 from object_detection.utils import ops as utils_ops
         18
         19 if StrictVersion(tf.__version__) < StrictVersion('1.12.0'):
         20     raise ImportError('Please upgrade your TensorFlow installation to v1.12.0+')
         21
```

Figure 3: Code of object detection tutorial

models are composed of multiple processing layers. Deep learning methods have proved to be very effective in speech recognition, visual object recognition, object detection and many other domains.

In the subsequent section, I will discuss a very basic method of object identification using TensorFlow. This experiment has been carried on Ubuntu 18.04.3 with Python, TensorFlow and Protobuf 3.9.

The following steps can be used for object detection using TensorFlow. This will identify objects kept in the *test_images* folder of the TensorFlow directory.

- Set up the environment; install TensorFlow and the Tensor GPU using the *pip* command. Pip is a tool for installing and managing Python packages. Pip 19 or later is required for TensorFlow 2.0.

```
python -m pip install tensorflow
```

```
python -m pip install tensorflow-gpu
```

Install the required object detection libraries, as follows:

```
# static compiler
pip install --user Cython
# backport of the contextlib module to
earlier Python versions.
pip install --user contextlib2
#python imaging library
pip install --user pillow
# For processing HTML and xml in python
pip install --user lxml
# open-source web application
pip install --user jupyter
#Plotting library for python
pip install --user matplotlib
```

Download Protocol Buffers (Protobuf). Run the following command from the *tensorflow/models/research/* folder:

```
protoc object_detection/protos/*.proto
--python_out=.
```

Or you may compile each file in the *protos* folder, individually; for example: